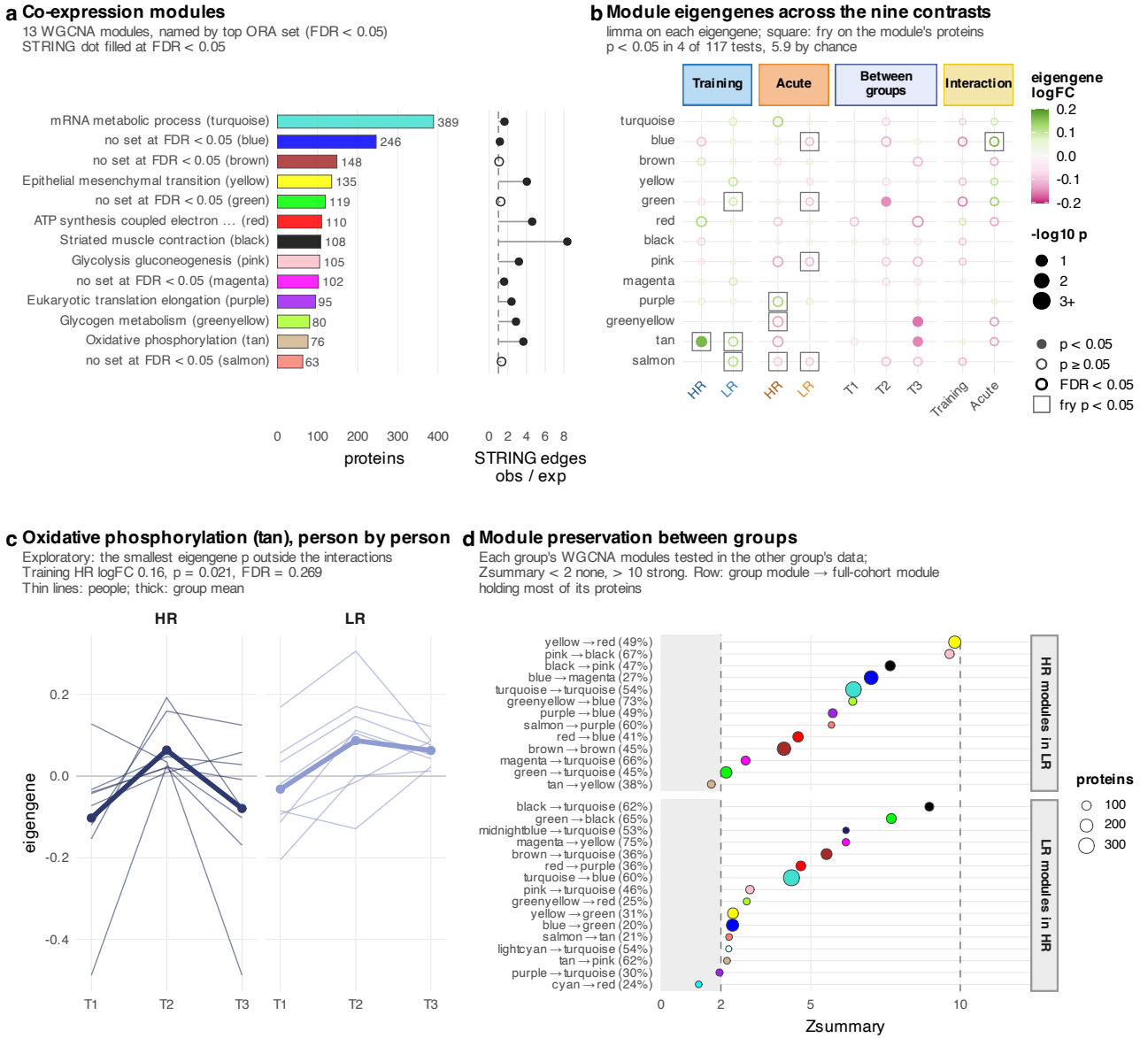
a Co-expression modules
13 WGCNA modules, named by top ORA set (FDR < 0.05)
STRING dot filled at FDR < 0.05
mRNA metabolic process (turquoise) 389
no set at FDR < 0.05 (blue) 246
no set at FDR < 0.05 (brown) 148
Epithelial mesenchymal transition (yellow) 135
no set at FDR < 0.05 (green) 119
ATP synthesis coupled electron … (red) 110
Striated muscle contraction (black) 108
Glycolysis gluconeogenesis (pink) 105
no set at FDR < 0.05 (magenta) 102
Eukaryotic translation elongation (purple) 95
Glycogen metabolism (greenyellow) 80
Oxidative phosphorylation (tan) 76
no set at FDR < 0.05 (salmon) 63
0 100 200 300 400
proteins
0 2 4 6 8
STRING edges obs / exp
b Module eigengenes across the nine contrasts
limma on each eigengene; square: fry on the module's proteins
p < 0.05 in 4 of 117 tests, 5.9 by chance
Training
Acute
Between groups
Interaction
turquoise
blue
brown
yellow
green
red
black
pink
magenta
purple
greenyellow
tan
salmon
HR LR HR LR T1 T2 T3 Training Acute
eigengene logFC
0.2
0.1
0.0
-0.1
-0.2
-log10 p
1
2
3+
p < 0.05
p ≥ 0.05
FDR < 0.05
fry p < 0.05
c Oxidative phosphorylation (tan), person by person
Exploratory: the smallest eigengene p outside the interactions
Training HR logFC 0.16, p = 0.021, FDR = 0.269
Thin lines: people; thick: group mean
HR
LR
eigengene
0.2
0.0
-0.2
-0.4
T1 T2 T3 T1 T2 T3
d Module preservation between groups
Each group's WGCNA modules tested in the other group's data;
Zsummary < 2 none, > 10 strong. Row: group module → full-cohort module
holding most of its proteins
yellow → red (49%)
pink → black (67%)
black → pink (47%)
blue → magenta (27%)
turquoise → turquoise (54%)
greenyellow → blue (73%)
purple → blue (49%)
salmon → purple (60%)
red → blue (41%)
brown → brown (45%)
magenta → turquoise (66%)
green → turquoise (45%)
tan → yellow (38%)
black → turquoise (62%)
green → black (65%)
midnightblue → turquoise (53%)
magenta → yellow (75%)
brown → turquoise (36%)
red → purple (36%)
turquoise → blue (60%)
pink → turquoise (46%)
greenyellow → red (25%)
yellow → green (31%)
blue → green (20%)
salmon → tan (21%)
lightcyan → turquoise (54%)
tan → pink (62%)
purple → turquoise (30%)
cyan → red (24%)
HR modules in LR
LR modules in HR
proteins
100
200
300
0 2 5 10
Zsummary